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Lifting More Than Weights: Multi-dimensional Health Benefits of Resistance Training in Children

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ABSTRACT

Recent advances in pediatric exercise physiology have opened new horizons in understanding the fundamental role of physical activity in children's physical, psychological, and cognitive development. In response to the alarming rise in childhood obesity, sedentary behavior, and the early onset of lifestyle-related non-communicable diseases, numerous studies have investigated the effects of structured exercise interventions, particularly resistance training (RT). Interventions involving RT can impact metabolic health, physical fitness, and cognitive performance in children and adolescents. Further, evidence from studies conducted between 2011 and 2025 indicates that RT, even in prepubertal populations, can safely enhance muscular strength, improve body composition, increase bone health, improve fundamental movement skill performance, reduce injury risk, and support cognitive functions. Moreover, tailoring training programs to individual needs within educational and community settings can promote long-term adherence to physical activity. Accordingly, RT can serve as a key strategy for improving child health in the 21st century.

Key words: Resistance Training, Pediatric Exercise Physiology, Physical Fitness, Metabolic Health, Cognitive Development, Injury Prevention

INTRODUCTION

Pediatric exercise physiology, as a dynamic branch of sports science, has seen remarkable advancements over the past decade, significantly expanding our understanding of the multi-dimensional effects of physical activity on children's physical, psychological, and cognitive development. These developments have been particularly important in addressing the growing prevalence of childhood obesity, declining physical activity levels, and the early emergence of lifestyle-related non-communicable diseases such as type 2 diabetes, metabolic syndrome, and cardiovascular disorders (World Health Organization, 2020; Moreno-Torres et al., 2025).

Systematic reviews and experimental studies from 2011 to 2025 have consistently demonstrated that regular physical activity offers benefits beyond (and independent from) obesity prevention prevention (Raiman et al., 2023). These include improvements in cardiorespiratory fitness, enhanced insulin sensitivity, healthier lipid profiles, stronger musculoskeletal function, improved executive brain functions, better academic performance, and enhanced psychosocial well-being (Biddle et al., 2019; Liu et al., 2024).

Among intervention strategies, resistance training (RT) and high-intensity interval training (HIIT) have emerged as effective methods for enhancing metabolic health and physical fitness, particularly in overweight or obese children (Zheng et al., 2025). Evidence suggests that RT can increase muscular strength, improve body composition, boost insulin sensitivity, and reduce visceral fat (García-Hermoso et al., 2020; Shen et al., 2025). Meta-analyses have reported moderate-to-large effects (SMD = 0.54–1.12) in improving strength, power, speed, agility, and sport-specific motor skills (Lesinski et al., 2020). In terms of safety, studies indicate that well-designed, supervised RT is safe even for prepubertal children, and that neuromuscular adaptations can enhance performance without negatively affecting growth plates (Ramirez-Campillo et al., 2023; Myers et al., 2017). These findings reinforce the importance of incorporating RT into school-based and community health programs.

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Personalized and Inclusive Approaches to Pediatric Training Design

Recent approaches to training design emphasize personalized interventions to meet the needs of children with chronic illnesses, neurodevelopmental disorders, or physical disabilities. This aligns with principles of personalized medicine and participatory rehabilitation, ensuring that training is tailored to each child's physiological and psychological characteristics (Rimmer et al., 2010; Bäßfuß et al., 2025). Such approaches have led to the development of multidisciplinary programs in school and community settings, aiming to foster long-term adherence to physical activity and the adoption of health-promoting behaviors from an early age (Hills et al., 2015; de Jesus, 2024).

Nevertheless, structural and psychosocial barriers remain. Major obstacles includes economic disparities, inadequate infrastructure, limited access to sports programs, and insufficient family or community support, particularly in underserved areas (Aubert et al., 2018; Sallis et al., 2012). Additionally, psychosocial determinants such as family dynamics, peer influence, intrinsic motivation, and mental health status play decisive roles in shaping motor behaviors and determining the success of interventions (Bauman et al., 2012; Fu, 2025).

A comprehensive health system–based approach integrating medical, behavioral, and environmental interventions is therefore essential to promote sustained and inclusive participation in children's physical activity. Grounded in recent scientific advancements, pediatric exercise physiology offers a robust framework for designing innovative prevention, intervention, and public health strategies for the 21st century.

Pediatric Benefits of Resistance Training

Early investigations in the 2010s primarily focused on RT's effects on muscular strength-fitness and body composition, reporting increased strength and reduced fat mass after 8–12 weeks of RT, performed at 60–80% of one-repetition maximum, for two to three sessions per week (García-Hermoso et al., 2020).

Pediatric Benefits of Resistance Training

International research over the past two decades on RT in children and adolescents has produced robust and extensive evidence regarding its impact on multiple dimensions of health. Systematic reviews and meta-analyses published between 2011 and 2025 have consistently demonstrated that RT can yield benefits beyond physical fitness, extending to mental health, cognitive performance, and fundamental motor skills (Moreno-Torres et al., 2025; Liu et al., 2024).

More recent research has broadened the scope to include bone health (Bäßfuß et al., 2025), injury prevention (Shen et al., 2025), and academic performance (Robinson, 2023). Notably, recent studies have targeted specific subgroups, such as overweight children, those with chronic illnesses, and elite young athletes. This targeted approach has generated more practical evidence applicable to educational policy and health program planning (de Jesus, 2024; Fu, 2025). Furthermore, research emphasizes multi-component programs combining RT with aerobic, skill-based, and balance training. These integrated models enhance physical literacy, reduce injury risk, and promote long-term athletic development (Hills et al., 2015; Lesinski et al., 2020).

Therefore, robust and extensive evidence have been accumulated, particularly over the past two decades, regarding RT impact on multiple dimensions of children and adolescents health. Systematic reviews and meta-analyses published between 2011 and 2025 have consistently demonstrated that RT can yield benefits beyond physical fitness, extending to mental health, cognitive performance, and fundamental motor skills (Moreno-Torres et al., 2025; Liu et al., 2024).

Resistance Training Protocols for Children, PEEP-FIE

The PEEP-FIE protocol (Program Initiation, Effective Intensity, Equipment Type, Progressive Overload, Frequency, Intervention Duration, and Exercise Combination) synthesises current scientific evidence and international guidelines to provide a structured framework for implementing RT in children and adolescents. It outlines key training elements that support safety, maximise health and performance outcomes, and promote long-term engagement in physical activity. Table 1 summarises these core components and their

Table 1. PEEP-FIE Protocol

Training Element	Description
Initiation	Begin with 1–2 sets of 8–15 repetitions at light-to-moderate intensity (30–60% 1RM) across 8–12 core resistance exercises.
Intensity and volume	6–8 exercises 60–80% 1RM, 2–3 sets of 8–15 repetitions for to achieve optimal strength and performance gains (Moreno-Torres et al., 2025).
Equipment	Combination* of equipment, including (but not limited to) free weights, weight machines, resistance bands, bodyweight exercises (i.e., plyometrics).
Progressive overload	Gradually increase training load, ensure correct technique, allow adequate rest between sets, and maintain direct supervision by qualified trainers (Myers, 2017).
Frequency	2–3 non-consecutive sessions per week, each lasting 40–60 minutes (García-Hermoso et al., 2020).
Duration	Typically 8–20 weeks; longer programs (up to 52 weeks) demonstrate more sustained results (Shen et al., 2025).
Exercise combination	Integrate RT with skill-based, balance, and plyometric training to support comprehensive motor development (Liu et al., 2024).

1RM: one-repetition maximum;
RT: resistance training.
*: different relative distribution of equipment (e.g. from total volume, 60% with free weights and 40% with weight machines) may produce greater strength gains in some participants (Lesinski et al., 2020)

Table 2. Resistance training: benefits, myths, and evidence

Key Benefit	Common Myth	Evidence/Reality
Enhancement of fundamental motor skills and athletic performance	Strength training makes kids bulky and slow.	RT improves speed, agility, balance, and coordination without excessive muscle gain in children (Lesinski et al., 2020; Shen et al., 2025).
Injury prevention	Weight training is dangerous for children and will lead to injuries.	Supervised RT reduces injury risk by strengthening muscles, joints, and connective tissues (Shen et al., 2025; García-Hermoso et al., 2020).
Safety, bone health, skeletal growth and growth plates	Lifting weights stunts children's growth by damaging growth plates.	Correct technique and qualified supervision make RT safe, improves bone mineral density, and may protect growth plates (Myers, 2017; Bäßfuß et al., 2025).
Physical literacy development	Strength training is only for athletes, not for everyday kids.	RT benefits all children, building skills and confidence for lifelong activity (Hills et al., 2015; de Jesus, 2024).
Fitness and long-term athlete development	Only playing sport improves fitness and performance.	Structured RT accelerates performance gains beyond sport practice alone (Lesinski et al., 2020; Shen et al., 2025).
Flexibility and balance	Strength training makes you stiff and less flexible.	RT combined with mobility or neuromuscular work can increase flexibility and balance (Moreno-Torres et al., 2025; Liu et al., 2024).
Retention and detraining	Once you're strong, you stay strong forever.	Strength declines quickly without ongoing training; maintenance programs sustain gains (Myers, 2017; Robinson, 2023).
Eccentric training and recovery	Soreness after training means it was a good workout.	DOMS is not always a marker of effectiveness; eccentric RT can reduce soreness and speed recovery (Fu, 2025; Moreno-Torres et al., 2025).
Cardiometabolic health	Cardio is the only way to improve heart health and metabolism.	RT improves lipid profiles and insulin sensitivity, even without major body composition changes (Liu et al., 2024; García-Hermoso et al., 2020).
Mental health and cognition	Strength training only helps the body, not the mind.	RT enhances brain function, focus, and emotional well-being via neurochemical changes (Fu, 2025; Robinson, 2023).

1RM: one-repetition maximum;
DOMS: delayed onset of muscle soreness.
RT: resistance training

recommended parameters, offering a practical reference for educators, coaches, and health professionals.

Analysis of RT Effects on Key Indicators of Growth and Health – Debunking Myths

Studies from 2010 to 2025 have assessed RT's influence on a wide range of physical, psychological, and cognitive factors in children. Meta-analyses confirm RT's role as a multi-dimensional intervention for improving youth health outcomes (Moreno-Torres et al., 2025; Liu et al., 2024). Despite the robust research to-date, myths are still prevalent (see Table 2).

Evidence Overview and Reflections

Robust evidence supports RT association with substantial physical, psychological, and cognitive benefits for children and adolescents, extending beyond growth-related changes. Benefits include improvements in muscular strength, body composition, bone health, athletic performance, flexibility, cardiometabolic health, and cognitive function (Moreno-Torres et al., 2025; Liu et al., 2024). Supervised RT may aid in injury prevention, which can lower medical costs, reduce time lost from school and sports, and improve overall participation rates (Shen et al., 2025). Additionally, RT's impact on physical literacy highlights its importance in fostering intrinsic motivation for physical activity, aligning with long-term athlete development principles (Hills et al., 2015; Lesinski et al., 2020). Physiological evidence suggests RT enhances neurocognitive outcomes via increased brain-derived neurotrophic factor levels, improving executive function and academic performance (Fu, 2025; Robinson, 2023). However, research challenges remain, particularly related to the variability in study design and protocols (limited comparability), and most research focuses on healthy populations, and long-term maintenance strategies are underexplored (Bäßfuß et al., 2025; Myers, 2017).

CONCLUSION

When properly designed and supervised, RT is safe and can enhance multiple health domains in children and adolescents. Beyond enhancing strength and sports performance, RT supports bone health, reduces risk of injuries, boosts physical literacy, and strengthens cognitive outcomes. To maximise benefits, RT programs should begin gradually at young ages, combine with aerobic, skill-based, and balance training, be implemented in schools and community settings with accessible equipment, and be tailored for special populations. Given the breadth of evidence, integration of RT into public health strategies is a must, shifting the perception of RT from a specialized or optional activity for adults to a foundational component of children's and young people's health promotion in the 21st century.

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